242695: hornblende metamonzonite, Olivers Well

(Cagacaroon Syenogranite, Rothsay Suite, Annean Supersuite, Youanmi Terrane, Yilgarn Craton)

Location and sampling

Yalgoo (SH 50-2), Mellenbye (2140)
MGA Zone 50, 436533E 6830669N

Sampled on 6 May 2019

This sample was collected from low outcrop west of the Cagacaroon Hills, about 5.8 km south of Five Mile Well, 4.2 km west-northwest of Gearless Well, and 1.8 km north-northeast of Olivers Well.

Geological context

The unit sampled is a hornblende metamonzonite of the Cagacaroon Syenogranite of the 2763–2742 Ma Rothsay Suite, near the northwestern margin of the Gullewa greenstone belt, west of the Yalgoo Dome (Zibra et al., 2018). The Rothsay Suite consists mainly of metagranodiorite to metatonalite, and is a component of the 2825–2733 Ma Annean Supersuite (Van Kranendonk et al., 2013). A quartz syenite from this locality yielded an igneous crystallization age of 2747 ± 3 Ma (sample 99969142, Fletcher and McNaughton, 2002). A metagranodiorite about 13.2 km to the northeast, within the Edamurta greenstone belt, yielded an igneous crystallization age of 2741 ± 3 Ma (GSWA 227224, Wingate et al., 2021).

Petrographic description

This sample is a medium-grained, inequigranular, hornblende metamonzonite (Fig. 1), consisting of about 40% K-feldspar, 20–25% plagioclase, 20% amphibole, 5–7% quartz, 5% biotite, 2–3% epidote, 2% magnetite, <1% titanite, and accessory apatite and zircon. K-feldspar is subhedral, tabular, up to 7 mm long, and exhibits microperthitic textures and some microcline twinning. Subhedral tabular plagioclase is up to 2.5 mm long and has been albitized and sericitized. Some feldspars consist of plagioclase relics within K-feldspar, which suggests potassic metasomatism has occurred. Green hornblende forms subhedral to euhedral crystals up to 2 mm in size, mostly in aggregates with minor biotite, magnetite and titanite, and is mostly replaced by pale green actinolite. Quartz is anhedral, irregular, interstitial to feldspars, and up to 1.5 mm in size. Most quartz exhibits undulose extinction, as well as minor development of sub-grains and recrystallization. Greenish brown biotite forms subhedral, equant laths up to 1 mm long, locally replaced by chlorite. Anhedral, irregular, pale yellow epidote is mostly associated with amphibole-rich aggregates. Magnetite forms anhedral to subhedral, equant grains up to 0.25 mm in size, and occurs in aggregates within amphiboles. Partially resorbed, subhedral titanite is up to 1.5 mm long, and apatite is mainly euhedral and up to 0.2 mm long. The monzonite lacks a tectonic fabric, although evidence of strain within quartz and K-feldspar suggests deformation occurred as moderate metamorphic grade.

Zircon morphology

Zircons isolated from this sample are colourless to dark brown and subhedral to euhedral. The crystals are up to 400 µm long, and equant to elongate, with aspect ratios up to 8:1. In cathodoluminescence (CL) images, concentric zoning is ubiquitous, some crystals exhibit sector zoning, and some are dominated by high-U, metamict zones. A CL image of representative zircons is shown in Figure 2.

Analytical details

This sample was analysed on 21–22 October 2021, using SHRIMP-B. Fifteen analyses of the M257 standard were obtained during the session, of which 11 analyses indicated an external spot-to-spot (reproducibility) uncertainty of 0.80% (1σ) and a $^{238}U/^{206}Pb^*$ calibration uncertainty of 0.32% (1σ). Calibration uncertainties are included in the errors of $^{238}U/^{206}Pb^*$ ratios and dates listed in Table 1. Common-Pb corrections were applied to all analyses using contemporaneous isotopic compositions determined according to the model of Stacey and Kramers (1975).

Results

Seventeen analyses were obtained from 17 zircons. Results are listed in Table 1, and shown in a concordia diagram (Fig. 3).

Interpretation

The analyses are concordant to slightly discordant (Fig. 3), and form a single group, based on their $^{207}Pb^*/^{206}Pb^*$ ratios.

Group I comprises 17 analyses (Table 1), which yield a weighted mean $^{207}Pb^*/^{206}Pb^*$ date of 2754 ± 4 Ma (MSWD = 0.85).

The date of 2754 ± 4 Ma for the 17 analyses in Group I is interpreted as the magmatic crystallization age of the monzonite.

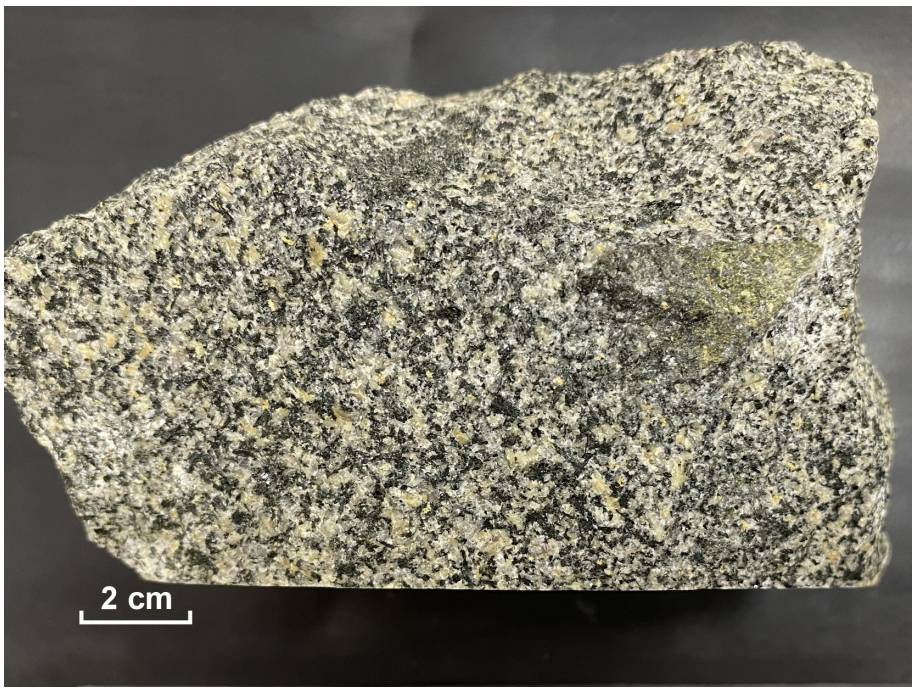

Figure 1. **Hand specimen image of sample 242695: hornblende metamonzonite, Olivers Well**

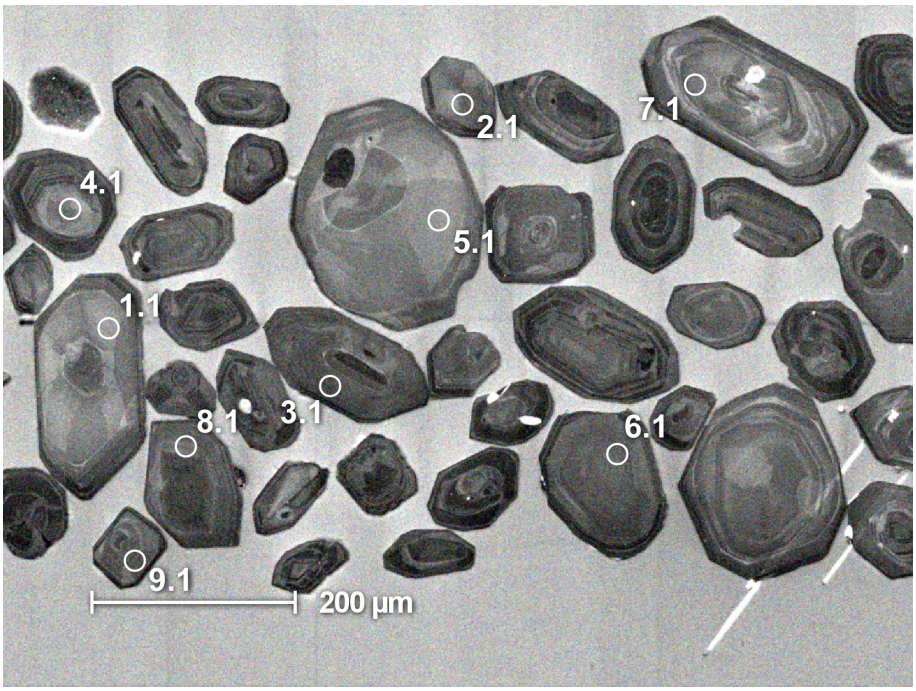

Figure 2. **Cathodoluminescence image of representative zircons from sample 242695: hornblende metamonzonite, Olivers Well. Numbered circles indicate the approximate locations of analysis sites**

Table 1. Ion microprobe analytical results for zircons from sample 242695: hornblende metamonzonite, Olivers Well

Group ID	Spot no.	Grain. spot	^{238}U (ppm)	^{232}Th (ppm)	$^{232}Th/^{238}U$	f204 (%)	$^{238}U/^{206}Pb$ ± 1σ		$^{207}Pb/^{206}Pb$ ± 1σ		$^{238}U/^{206}Pb^*$ ± 1σ		$^{207}Pb^*/^{206}Pb^*$ ± 1σ		$^{238}U/^{206}Pb^*$ date (Ma) ±1σ		$^{207}Pb^*/^{206}Pb^*$ date (Ma) ±1σ		Disc. (%)
I	8	8.1	215	185	0.86	0.000	1.895	0.024	0.18958	0.00099	1.895	0.024	0.18958	0.00099	2732	29	2739	9	0.3
I	15	15.1	236	216	0.92	0.061	1.935	0.023	0.19058	0.00098	1.937	0.023	0.19003	0.00101	2684	27	2742	9	2.1
I	10	10.1	205	211	1.03	0.014	1.897	0.023	0.19024	0.00105	1.897	0.023	0.19011	0.00106	2730	28	2743	9	0.5
I	7	7.1	142	147	1.03	0.052	1.917	0.025	0.19068	0.00115	1.918	0.025	0.19022	0.00118	2705	29	2744	10	1.4
I	2	2.1	248	282	1.14	-0.010	1.864	0.021	0.19084	0.00088	1.864	0.021	0.19093	0.00089	2769	26	2750	8	-0.7
I	13	13.1	165	160	0.97	0.074	1.943	0.024	0.19169	0.00108	1.944	0.024	0.19102	0.00112	2675	28	2751	10	2.8
I	4	4.1	240	293	1.22	-0.031	1.913	0.022	0.19092	0.00091	1.912	0.022	0.19120	0.00092	2712	26	2753	8	1.5
I	16	16.1	238	243	1.02	0.012	1.923	0.023	0.19163	0.00096	1.923	0.023	0.19152	0.00097	2699	26	2755	8	2.0
I	11	11.1	467	881	1.89	0.050	1.853	0.020	0.19206	0.00067	1.854	0.020	0.19161	0.00068	2781	25	2756	6	-0.9
I	17	17.1	276	353	1.28	0.020	1.898	0.022	0.19180	0.00088	1.898	0.022	0.19162	0.00089	2728	26	2756	8	1.0
I	14	14.1	243	236	0.97	0.046	1.893	0.022	0.19211	0.00095	1.894	0.022	0.19170	0.00098	2733	27	2757	8	0.9
I	9	9.1	244	291	1.19	0.000	1.845	0.022	0.19180	0.00093	1.845	0.022	0.19180	0.00093	2792	27	2758	8	-1.2
I	5	5.1	129	139	1.08	-0.040	1.824	0.025	0.19168	0.00125	1.824	0.025	0.19204	0.00128	2819	31	2760	11	-2.1
I	3	3.1	295	415	1.40	0.042	1.884	0.029	0.19251	0.00082	1.885	0.029	0.19213	0.00083	2744	35	2761	7	0.6
I	6	6.1	345	472	1.37	-0.007	1.937	0.021	0.19211	0.00074	1.937	0.021	0.19217	0.00074	2683	24	2761	6	2.8
I	12	12.1	182	239	1.31	0.040	1.929	0.023	0.19284	0.00102	1.930	0.023	0.19249	0.00104	2692	27	2764	9	2.6
I	1	1.1	117	120	1.03	0.022	1.923	0.026	0.19442	0.00212	1.923	0.026	0.19423	0.00213	2699	31	2778	18	2.9

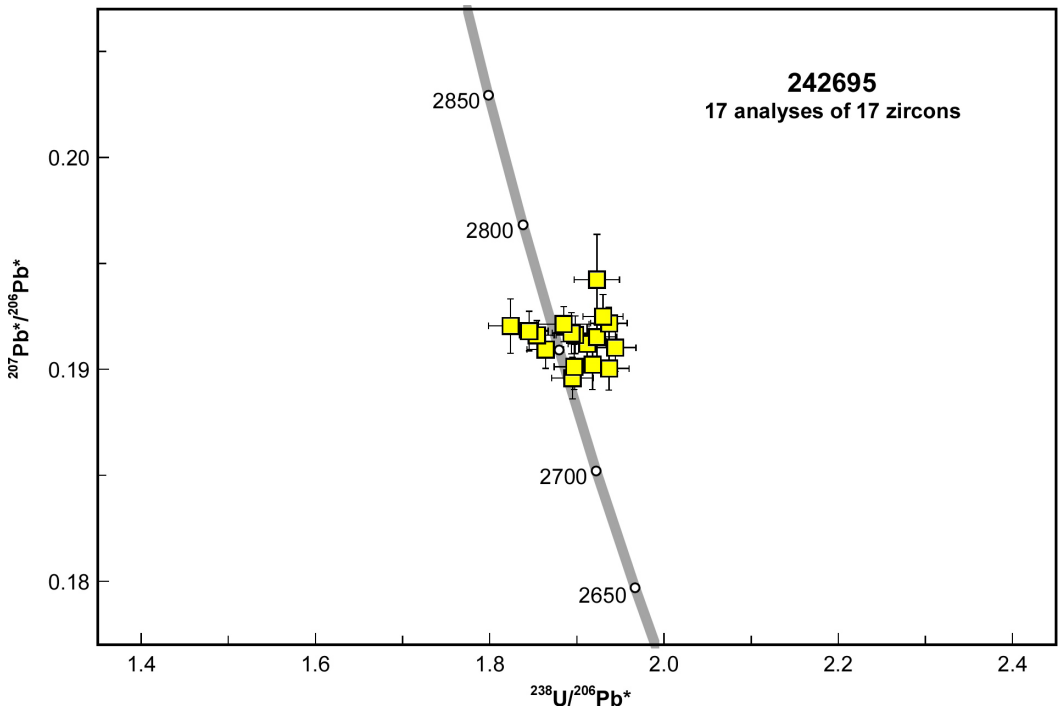

Figure 3. U–Pb analytical data for sample 242695: hornblende metamonzonite, Olivers Well. Yellow squares indicate Group I (magmatic zircons)

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Recommended reference for this publication

Wingate, MTD, Fielding, IOH, Lu, Y and Ivanic, TJ 2022, 242695: hornblende metamonzonite, Olivers Well; Geochronology Record 872: Geological Survey of Western Australia, 4p.

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